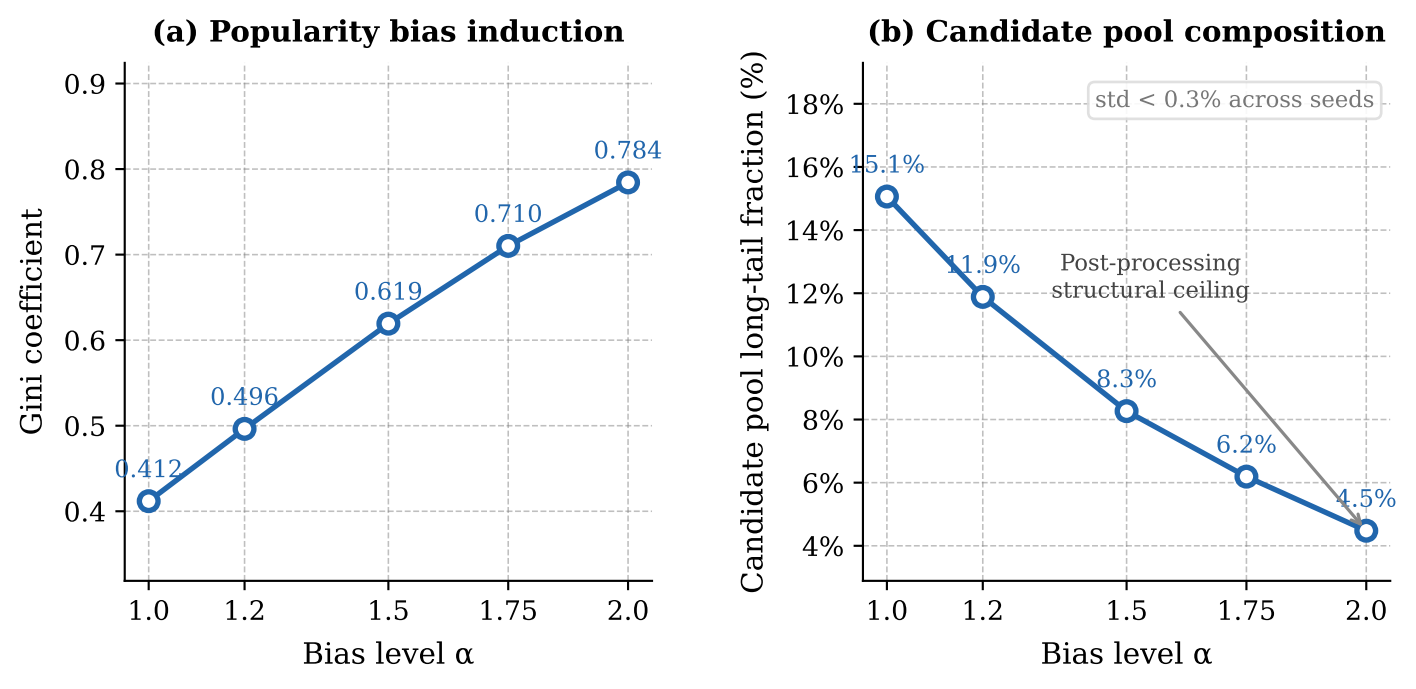
(a) Popularity bias induction
Gini coefficient
0.9
0.8
0.7
0.6
0.5
0.4
0.412
0.496
0.619
0.710
0.784
1.0
1.2
1.5
1.75
2.0
Bias level α
(b) Candidate pool composition
Candidate pool long-tail fraction (%)
18%
16%
14%
12%
10%
8%
6%
4%
std < 0.3% across seeds
15.1%
11.9%
8.3%
6.2%
4.5%
Post-processing
structural ceiling
1.0
1.2
1.5
1.75
2.0
Bias level α